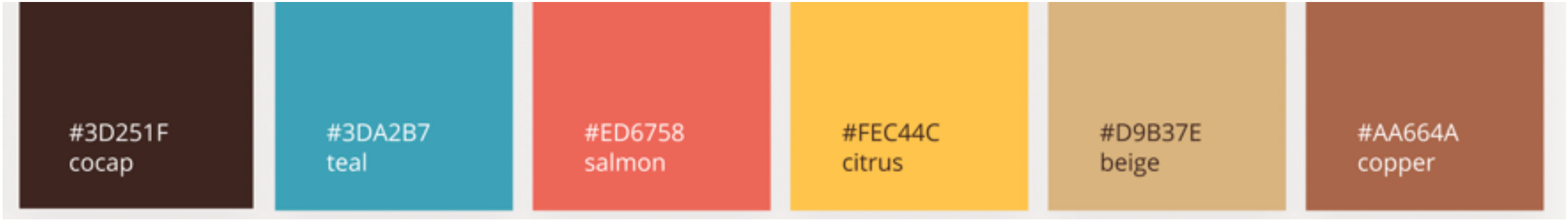


$$\alpha \equiv \mathcal{N}(\mu_\alpha, \Sigma_\alpha)$$

$$\beta = \mathcal{N}(\mu_\beta, \Sigma_\beta)$$

Property: An affine transformation of a Gaussian r.v. is also Gaussian



#3D251F
cocap

#3DA2B7
teal

#ED6758
salmon

#FEC44C
citrus

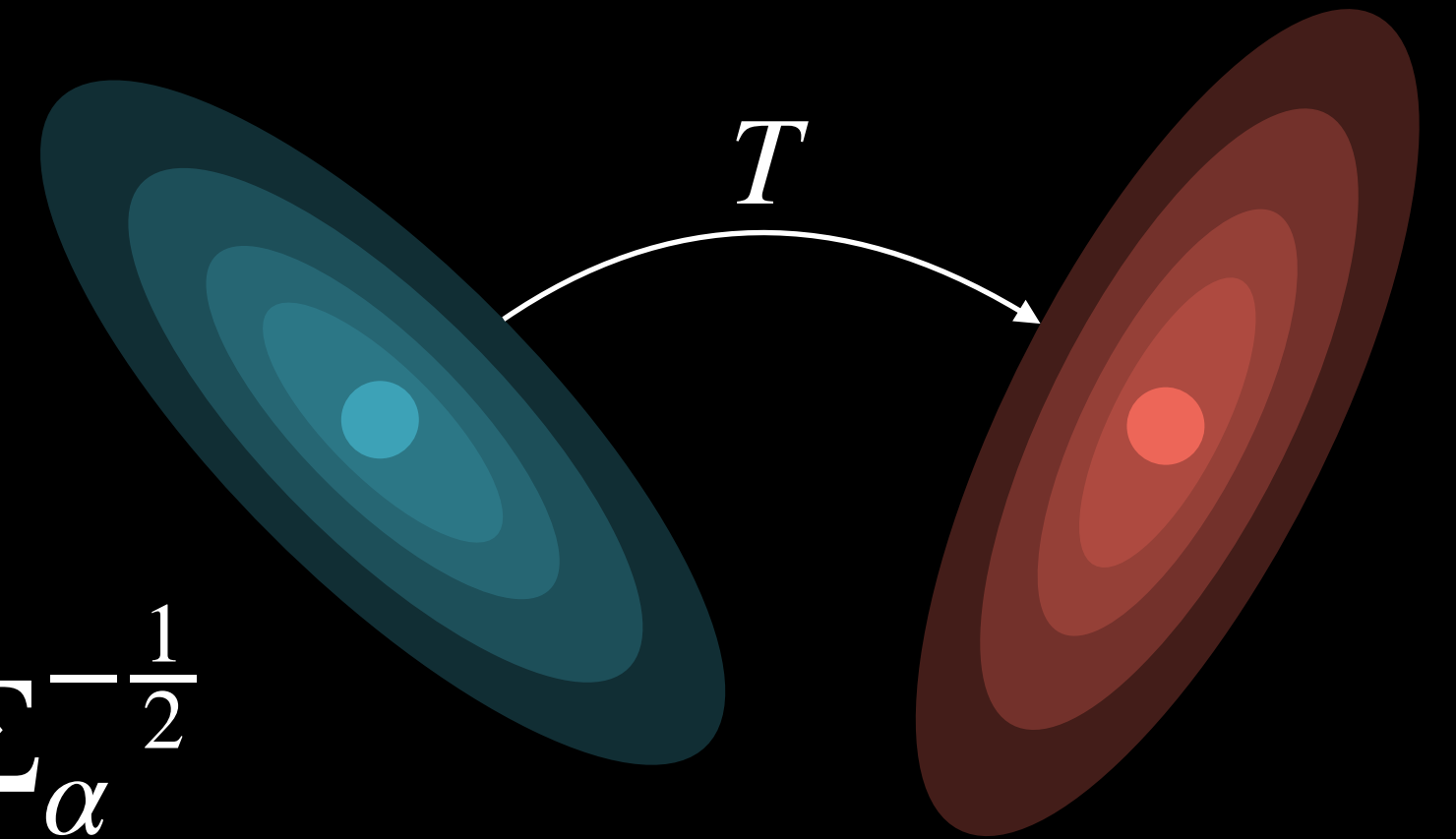
#D9B37E
beige

#AA664A
copper

OT between N-D Gaussians

$$\alpha = \mathcal{N}(\mu_\alpha, \Sigma_\alpha)$$

$$\beta = \mathcal{N}(\mu_\beta, \Sigma_\beta)$$



$$T(x) = \mu_\beta + A(x - \mu_\alpha) \quad A = \Sigma_\alpha^{-\frac{1}{2}} (\Sigma_\alpha^{\frac{1}{2}} \Sigma_\beta \Sigma_\alpha^{\frac{1}{2}})^{\frac{1}{2}} \Sigma_\alpha^{-\frac{1}{2}}$$

Proof:

1. Show that $T(x) \sim \mathcal{N}(\mu_\beta, \Sigma_\beta)$

2. Note that $T = \nabla \varphi$ for $\varphi(x) = \frac{1}{2} \langle x - \mu_\alpha, A(x - \mu_\alpha) \rangle + \langle \mu_\beta, x \rangle$

Property: An affine transformation of a Gaussian r.v. is also Gaussian

By Brenier's theorem, T is the optimal transport map

Monge-Ampère Equation